DAY_01

Duration: 2 hours

1. Create tables and insert records using INSERT statements.
2. Insert multiple records into a table.
3. Update specific records using UPDATE and WHERE clauses.
4. Delete selected records using DELETE statement.
5. Delete all records from a table using DELETE.
6. Use TRUNCATE to remove all rows from a table.
7. Compare the behavior of DELETE and TRUNCATE commands.

DAY_02

Duration: 2 hours

1. Defining Data Types for Columns
2. Creating, Altering, Dropping Constraints
3. Aggregate Functions: SUM(), AVG(), COUNT(), MAX(), MIN(), COUNT(), Group By, Having Clause
4. Using Like, Distinct, Order By, Between...And
5. Comparing Nulls, Using IN/Not-In

DAY_03

Duration: 2 hours

1. Write queries using ROW_NUMBER, RANK, DENSE_RANK to rank employees
2. Running totals and moving averages using SUM OVER with PARTITION BY
3. Rewrite complex subqueries as CTEs for readability; recursive CTE for org hierarchy

DAY_04

Duration: 2 hours

i. Defining Data Types for Columns
ii. Creating, Altering, Dropping Constraints
iii. Aggregate Functions: SUM(), AVG(), COUNT(), MAX(), MIN(), COUNT(), Group By, Having Clause
iv. Using Like, Distinct, Order By, Between...And
v. Comparing Nulls, Using IN/Not-In

DAY_05

Duration: 2 hours

i. Union/Union ALL Queries on Various type of Joins using OLD and SQL Standard Syntax
ii. Copying table structure, Copying data from one table to another
iii. Using AUTO_INCREMENT

DAY_06

Duration: 2 hours

i. Write queries using ROW_NUMBER, RANK, DENSE_RANK to rank employees
ii. Running totals and moving averages using SUM OVER with PARTITION BY
iii. Rewrite complex subqueries as CTEs for readability; recursive CTE for org hierarchy

DAY_07

Duration: 2 hours

i. Subqueries, Correlated Queries
ii. Using Exists/Not-Exists
iii. Using Commit/Rollback/Savepoint
iv. Granting/revoking privileges on database objects
v. Creating Views, Querying using Views
vi. Creating Indexes
vii. Creating Temporary Tables

DAY_08

Duration: 2 hours

i. Use of flow control statement in Stored Procedure
ii. Use of conditional statements in Stored Procedure

DAY_09

Duration: 2 hours

i. Create Before Triggers and After Triggers Creating Temporary Tables

DAY_10

Duration: 2 hours

i. Exception handling in Stored Procedure
ii. Exception handling with various handler actions

DAY_11

Duration: 2 hours

i. Subqueries, Correlated Queries
ii. Using Exists/Not-Exists
iii. Using Commit/Rollback/Savepoint
iv. Granting/revoking privileges on database objects
v. Creating Views, Querying using Views
vi. Creating Indexes
vii. Creating Temporary Tables

DAY_12

Duration: 2 hours

i. Features of NoSQL Database
ii. Structured vs. Semi-structured and Unstructured Data
iii. Difference between RDBMS and NoSQL databases
iv. CAP Theorem, BASE Model
v. Categories of NoSQL Databases
    A. Key-Value Store
    B. Document Store
    C. Column-Oriented
    D. Graph
vi. Introduction to MongoDB, Features of MongoDB
vii. MongoDB command interface and MongoDB compass
viii. MongoDB Documents & Collections
ix. RDBMS & MongoDB analogies relations/tables => collections; tuples/records => documents JSON and BSON documents

DAY_13
Duration: 2 hours

i. Using MongoDB Shell and Compass
ii. Creating database, connecting to a database, Creating Collections

DAY_14
Duration: 2 hours

i. MongoDB: Complex Read Using Operators, Sorting Operations, Creating Indexes
ii. Performing CRUD operations: insert, find with filters, update, delete

DAY_15
Duration: 2 hours

i. Complex read queries using comparison and logical operators with projections
ii. Sorting, pagination (skip/limit), UPSERT